1Small Office LAN Setup (2 Computers) – Practical Answer

Aim:
To create a LAN using a switch, connect two PCs, assign IP addresses, and verify connectivity using ping.

Requirements:
Two computers, one network switch, two Ethernet cables.

Procedure:
First, connect both computers to the switch using Ethernet cables and ensure the switch is powered on. Next, assign IP addresses to both computers. In the first computer, open Network Settings, go to IPv4 properties, and set the IP address as 192.168.1.1 and subnet mask as 255.255.255.0. In the second computer, assign IP address 192.168.1.2 with the same subnet mask 255.255.255.0.

After configuring IP addresses, open Command Prompt on the first computer and type “ping 192.168.1.2” to check connectivity. Similarly, from the second computer, type “ping 192.168.1.1”. If replies are received, the connection is successful.

Result:
The LAN is successfully established between the two computers, and communication is verified using the ping command.

2 Home Network Troubleshooting (2 PCs) – Practical Answer

Aim:
To troubleshoot a network issue between two computers by configuring correct IP addresses and verifying connectivity using the ping command.

Requirements:
Two computers, Ethernet cables (or switch), network interface.

Procedure:
First, check the physical connections between both computers and ensure cables are properly connected and network ports are active.

Next, check the IP configuration on both systems. Open Command Prompt and type ipconfig to view current IP addresses. If the IP addresses are incorrect or not in the same network, reconfigure them.

Go to Network Settings → Change Adapter Settings → Right-click on Ethernet → Properties → Internet Protocol Version 4 (IPv4). Select “Use the following IP address” and assign proper IP addresses.

For the first computer, set:
IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0

For the second computer, set:
IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0

After assigning IP addresses, open Command Prompt on the first computer and type ping 192.168.1.2. Then, from the second computer, type ping 192.168.1.1.

If replies are received, the network issue is resolved.

Result:
The network problem is successfully identified and fixed by assigning correct IP addresses, and connectivity between both computers is verified using the ping command.

3 **Data Transfer Visualization – Practical Answer**

Aim:
To create a network and observe how data packets travel between two computers using simulation mode.

Requirements:
Two PCs, one switch, Ethernet cables, network simulation software (e.g., Cisco Packet Tracer).

Procedure:
First, open Cisco Packet Tracer and create a new workspace. Add two PCs and one switch from the device list.

Next, connect both PCs to the switch using copper straight-through cables. Ensure all connections are properly established.

Now, assign IP addresses to both PCs. Click on PC1, go to Desktop → IP Configuration, and set the IP address as 192.168.1.1 with subnet mask 255.255.255.0. Then click on PC2 and assign IP address 192.168.1.2 with the same subnet mask.

After configuration, switch to **Simulation Mode** from the bottom panel. Select the **Add Simple PDU (packet)** tool and click on PC1, then click on PC2 to send a packet.

Observe the packet movement step-by-step. Use the **Capture/Forward** button to see how the packet travels from PC1 to the switch and then to PC2. You can also view details like ARP request, ICMP packet, and acknowledgements.

If needed, you can also open Command Prompt on PC1 and type ping 192.168.1.2 to generate continuous packet flow in simulation mode.

Result:
The network is successfully created, and the flow of data packets between two computers is observed using simulation mode.

4 **Website Access Simulation – Practical Answer**

Aim:
To create a simple network with a server and observe HTTP communication by capturing packets.

Requirements:
One PC, one server, one switch, Ethernet cables, network simulation software (e.g., Cisco Packet Tracer).

Procedure:
First, open Cisco Packet Tracer and create a new workspace. Add one PC, one server, and one switch from the device list.

Next, connect the PC and server to the switch using copper straight-through cables. Ensure all devices are properly connected.

Now, assign IP addresses. Click on the PC, go to Desktop → IP Configuration, and set IP address as 192.168.1.2 with subnet mask 255.255.255.0. Then click on the server and assign IP address 192.168.1.1 with the same subnet mask.

After that, configure the server. Click on the server, go to Services → HTTP, and turn the HTTP service ON. You can edit the default webpage if required.

Next, switch to **Simulation Mode**. Open the web browser on the PC (Desktop → Web Browser) and type the server IP address http://192.168.1.1 in the address bar.

Observe the packet flow in simulation mode. Use the **Capture/Forward** option to see how HTTP request and response packets travel between the PC and server. You can also check details such as DNS (if used), TCP handshake, and HTTP communication.

Result:
The website is successfully accessed from the PC, and HTTP packet transfer between the client and server is observed using simulation mode.

5 **Protocol Identification Task – Practical Answer (Easy Language)**

Aim:
To simulate communication and identify Ethernet, IP, TCP, and HTTP protocols using Cisco Packet Tracer.

Requirements:
One PC, one server, one switch, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a new project. Add one PC, one server, and one switch.

Next, connect the PC and server to the switch using straight-through cables. Make sure all devices are properly connected.

Now, assign IP addresses. Click on the PC → Desktop → IP Configuration and set IP address as 192.168.1.2 and subnet mask 255.255.255.0. Then click on the server and assign IP address 192.168.1.1 with the same subnet mask.

After that, enable HTTP service on the server. Click on server → Services → HTTP → Turn ON.

Now, go to **Simulation Mode**. Open the web browser on the PC and type http://192.168.1.1 to access the website.

Observe the packets moving between PC and server. Click on the moving packet and open **PDU Details**. Here you can see different protocol layers:

- **Ethernet Layer** – shows MAC address (data link layer)
- **IP Layer** – shows IP address (network layer)
- **TCP Layer** – shows port numbers and connection (transport layer)
- **HTTP Layer** – shows request and response (application layer)

Use **Capture/Forward** to see each step clearly.

Result:
The communication is successfully simulated, and Ethernet, IP, TCP, and HTTP protocols are identified in different layers.

6 **Network Monitoring Scenario – Practical Answer (Easy Language)**

Aim:
To capture and observe source/destination IP addresses and port numbers during data transfer.

Requirements:
One PC, one server, one switch, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a new network. Add one PC, one server, and one switch.

Next, connect the PC and server to the switch using straight-through cables. Ensure proper connections.

Now, assign IP addresses. Click on PC → Desktop → IP Configuration and set IP address as 192.168.1.2 and subnet mask 255.255.255.0. Then click on the server and set IP address as 192.168.1.1 with the same subnet mask.

After that, enable HTTP service on the server by going to Services → HTTP → Turn ON.

Now switch to **Simulation Mode**. Open the web browser on the PC and type http://192.168.1.1 to start communication.

Observe the packet movement. Click on the packet and open **PDU Details**.

In the details, note the following:

- **Source IP Address:** 192.168.1.2 (PC)
- **Destination IP Address:** 192.168.1.1 (Server)
- **Source Port Number:** Random (e.g., 1025)
- **Destination Port Number:** 80 (HTTP)

Use **Capture/Forward** to see each step of communication clearly.

Result:
The data transfer is successfully monitored, and source/destination IP addresses and port numbers are identified.

7 **Full Network Connectivity Test – Practical Answer (Easy Language)**

Aim:
To test communication between all devices in a network and ensure full connectivity using ping.

Requirements:
Multiple PCs, switch, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a network. Add multiple PCs (for example PC1, PC2, PC3) and one switch.

Next, connect all PCs to the switch using straight-through cables. Make sure all connections are proper and active.

Now, assign IP addresses to all PCs in the same network:

- PC1: 192.168.1.1
- PC2: 192.168.1.2
- PC3: 192.168.1.3
  Subnet Mask for all: 255.255.255.0

After that, open Command Prompt on each PC.

From PC1, type:
ping 192.168.1.2
ping 192.168.1.3

From PC2, type:
ping 192.168.1.1
ping 192.168.1.3

From PC3, type:
ping 192.168.1.1
ping 192.168.1.2

Check the results. If you see “Reply from…”, it means devices are connected properly. If not, check IP address and cable connections.

Result:
All devices in the network are successfully connected, and full connectivity is verified using the ping command.

8 **Branch Office Connection – Practical Answer (Easy Language)**

Aim:
To connect two branch offices using subnets and routers and ensure communication between both networks.

Requirements:
Two routers, two switches, multiple PCs, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a new network. Add two routers (R1 and R2), two switches, and PCs for both branches.

Next, connect devices:

- Connect PCs of Branch 1 to Switch 1
- Connect PCs of Branch 2 to Switch 2
- Connect Switch 1 to Router R1
- Connect Switch 2 to Router R2
- Connect Router R1 to Router R2 using a serial or Ethernet cable

Now, create two different subnets:

- **Branch 1 Network:** 192.168.1.0 /24
- **Branch 2 Network:** 192.168.2.0 /24

Assign IP addresses:

☞ **Branch 1 PCs:**

- PC1: 192.168.1.2
- PC2: 192.168.1.3
- Default Gateway: 192.168.1.1

☞ **Router R1 (connected to Branch 1):**

- IP Address: 192.168.1.1

☞ **Branch 2 PCs:**

- PC3: 192.168.2.2
- PC4: 192.168.2.3
- Default Gateway: 192.168.2.1

☞ **Router R2 (connected to Branch 2):**

- IP Address: 192.168.2.1

Now configure connection between routers:

- On R1, assign IP: 10.0.0.1
- On R2, assign IP: 10.0.0.2

After that, configure routing:

☞ On Router R1:

- Add route to Branch 2 network:
  ip route 192.168.2.0 255.255.255.0 10.0.0.2

☞ On Router R2:

- Add route to Branch 1 network:
  ip route 192.168.1.0 255.255.255.0 10.0.0.1

Finally, test connectivity:

- From PC in Branch 1, ping PC in Branch 2
  Example: ping 192.168.2.2

If you get reply, communication is successful.

Result:
Both branch networks are successfully connected using routers and subnets, and communication between them is verified.

9 **IPv6 Implementation – Practical Answer (Easy Language)**

Aim:
To configure IPv6 addresses on devices and test connectivity between them.

Requirements:
Two PCs, one switch, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a new network. Add two PCs and one switch.

Next, connect both PCs to the switch using straight-through cables. Make sure all connections are proper.

Now, assign IPv6 addresses to both PCs.

☞ On **PC1**:

- Go to Desktop → IP Configuration
- Under IPv6, enter:
    - IPv6 Address: 2001:db8:1::1
    - Prefix Length: 64

☞ On **PC2**:

- Go to Desktop → IP Configuration
- Enter:
    - IPv6 Address: 2001:db8:1::2
    - Prefix Length: 64

After configuring IPv6, open Command Prompt on PC1.
Type:
ping 2001:db8:1::2

Then on PC2, type:
ping 2001:db8:1::1

If you receive "Reply from…", it means devices are connected successfully.

Result:
IPv6 addresses are successfully configured, and connectivity between devices is verified using the ping command.

10 **Network Simulation Project – Practical Answer (Easy Language)**

Aim:
To create a simple network topology and test communication between nodes using Cisco Packet Tracer.

Requirements:
Multiple PCs, one switch, Ethernet cables, Cisco Packet Tracer.

Procedure:
First, open Cisco Packet Tracer and create a new project. Add 3 PCs (PC1, PC2, PC3) and one switch.

Next, connect all PCs to the switch using straight-through cables. Ensure all devices are properly connected and links are active.

Now, assign IP addresses to each PC:

- PC1: 192.168.1.1
- PC2: 192.168.1.2
- PC3: 192.168.1.3
  Subnet Mask for all: 255.255.255.0

After assigning IP addresses, open Command Prompt on each PC.

From PC1, type:
ping 192.168.1.2
ping 192.168.1.3

From PC2, type:
ping 192.168.1.1
ping 192.168.1.3

From PC3, type:
ping 192.168.1.1
ping 192.168.1.2

Check the results. If “Reply from…” is shown, communication is successful.

You can also switch to **Simulation Mode** to see how packets travel between nodes step-by-step.

Result:
The simple network topology is successfully created, and communication between all nodes is verified.